

UTKARSH POPHLI

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SUMMARY

Machine Learning engineer with 2+ years taking AI/ML and Generative AI solutions from concept to production. Hands-on across traditional ML, deep learning, NLP, and LLM-based systems — model development, rigorous evaluation and scoring, and containerized deployment. Strong Python software development following object-oriented design patterns, grounded in linear algebra, probability, statistics, and optimization.

WORK EXPERIENCE

Fractal Analytics

Pune

Data Scientist

11/2025 – Present

- Reduced multi-agent pipeline latency by 40% across Fortune 500 deployments via a Temporal + LangGraph hybrid architecture, containerized with Docker and shipped through CI/CD gates; >99% uptime across 3 enterprise clients.
- Improved RAG retrieval precision from 61% to 84% with a chunking and retrieval scoring framework using offline regression checks and benchmark-driven evaluation, cutting hallucination incidents by 35%.
- Deployed enterprise GenAI services on Azure OpenAI Service and AKS with managed endpoints and deployment pipelines; handled model selection, prompt management, and structured output contracts.
- Authored ADRs and technical whitepapers on agentic AI platform architecture, contributing to engineering standards and reusable frameworks.

Green Rider Technology

Gurugram

Software Engineer—AI/ML

09/2025 – 11/2025

- Improved query resolution speed by 35% and cut agent fallback below 8% by building Conversational AI agents over unstructured support data, with rubric-based evaluation and human-escalation triggers.
- Accelerated developer productivity by 25% by architecting an enterprise coding assistant with a ReAct-style orchestration layer on Azure OpenAI Service, including retrieval quality evaluation and failure-mode instrumentation via distributed tracing.
- Cut prototype-to-production time by 30% and mentored engineers across three product teams via reusable LangGraph templates with CI/CD hooks and documented onboarding guides.

Siemens R&D

Goa

AI/ML Engineer

09/2024 – 09/2025

- Reduced internal document query time by 60% by architecting an enterprise RAG chatbot on AWS with retrieval scoring, chunking evaluation, and a runbook covering alerting, degradation handling, and incident response.
- Improved model accuracy by 25% and signal quality by 35% on a time series pipeline via EDA, feature engineering, and hyperparameter optimization; tracked experiments in MLflow and validated through offline regression tests and a Streamlit observability dashboard.
- Deployed containerized REST APIs on Azure with production monitoring and documented rollback procedures, resolving critical backend reliability issues on ElectrificationX.

EDUCATION

Madhav Institute of Technology & Science Gwalior

B.Tech in Internet of Things

2020 - 2024

CGPA: 8.32 / 10.0

SKILLS

Languages: Python, SQL

Machine Learning & AI: Machine Learning, Deep Learning, NLP, Named Entity Recognition (NER), Text Classification, LLMs, Generative AI, Time Series Modeling

Applied Mathematics: Linear Algebra, Probability, Statistics, Optimization, Numerical Methods

ML & DL Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, XGBoost, LightGBM, NumPy, Pandas, spaCy, OpenCV, Matplotlib, Plotly

AI Agent Development: LangChain, LangGraph, CrewAI, LlamaIndex, Microsoft Copilot Studio, Multi-Agent Orchestration, Orchestration Patterns, Tool Integrations, Agentic Memory, Temporal Workflows

LLM & GenAI Engineering: Prompt Engineering & Management, Model Selection, Structured Outputs, Retrieval-Augmented Generation (RAG) Pipelines, Chunking & Embeddings, Fine-Tuning, LoRA, QLoRA, RLHF, Guardrails, Responsible AI

Azure: Azure OpenAI Service, Azure AI Foundry, Azure ML, Azure AI Search, AKS, Azure Storage, Entra ID, Deployment Pipelines

AWS & GCP: Bedrock, SageMaker, Lambda, S3, Vertex AI, Agent Engine

MLOps & LLMOps: Docker & Dockerfiles, Kubernetes, MLflow (tracking, evaluation & scoring), Git, CI/CD Pipelines, Model Monitoring, Self-Hosted Inference (vLLM, Ollama)

Model Evaluation & Scoring: MLflow, Automated & Human Evaluation, Benchmark Design, BLEU, ROUGE, Offline Regression Testing

Software Engineering: Python, Object-Oriented Design, Design Patterns, Asynchronous Programming, REST APIs, FastAPI, Automated Testing, Debugging

Data & Retrieval: PostgreSQL, MongoDB, SQL, Vector Databases (Pinecone, ChromaDB, FAISS), Semantic Search, Embeddings

Practices: Agile/Scrum, Code Review, ADRs, Technical Documentation, Mentoring

PROJECT EXPERIENCE

ML Algorithms from Scratch



- Implemented core ML formulations from scratch — gradient descent, matrix decompositions, probabilistic estimators — with interactive optimization-convergence visualizations.

GLOW: Windows-Agent



- Autonomous vision-based multi-agent OS workflow engine (Planner-Executor-Verifier) with 92 custom tools via PyQt6.

CERTIFICATIONS

• Claude Certified Architect — Anthropic

• Fractal Certified: LLMops Associate

• Oracle Cloud Certified Professional — Generative AI